

Unit 6 – Inference for Categorical Data: Proportions
Notes 4 – Conducting a Significance Test for Proportions

Important Significance Test Notes

- The p-value is a _____; it assumes the null hypothesis is true. Assuming that H_0 is true, the p-value measures how likely it is to get evidence for H_a (as “large” or “larger” in the direction of the alternative) by chance alone.
- You must name the test, either by writing down the test name (_____) or by writing down the generic z formula.
- For a hypothesis test, we check if _____ and for confidence intervals we check _____
- When there is not convincing evidence to reject H_0 , it does not mean there is evidence that H_0 is true or H_a is false.

Conducting a Significance Test

_____:

- Define p = in words
- Identify the hypotheses you are testing and write down your significance level

_____:

- You will be creating a sampling distribution, which requires you to check
 - Random (Random Sample or Random Assignment)
 - Independent (When sampling, check 10% condition)
 - Normal (Check large counts condition with $np_0 \geq 10$ and $n(1 - p_0) \geq 10$)

_____:

- Identify the inference you are performing: _____
- Calculate your test statistic:

- Find your p-value by calculating the probability of getting a z-statistic this extreme or more extreme in the direction of the alternative hypothesis. (On TI-84 use normalcdf)

_____:

Because our p-value of <value> is <bigger/smaller> than our significance level of <level>, we <reject or fail to reject H_0 >. We <do/don't> have convincing evidence that the population proportion <context of what this means in the problem>

Example: One Sample Z Test for a Proportion

Allan reads a report that says 80% of US high school students have a computer at home. He believes the proportion is smaller than 0.8 at our school. Allan chooses an SRS of 60 students and records whether they have a computer at home. He wants to test this at a 5% significance level. In his sample, 41 of the students have a computer at home.

a) Is this evidence that the proportion of students who have a computer at home at our school is less than the national proportion?

b) Check your solution using the TI-84:

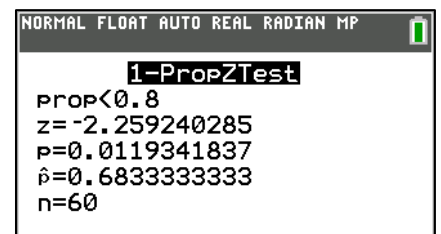
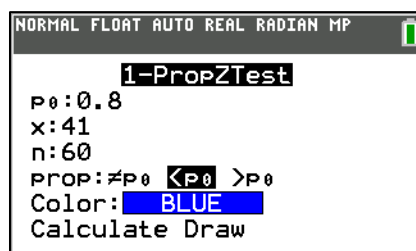
STAT → Tests → 5: 1-Prop Z Test

$p_0 =$

x:

n:

prop: $\neq p_0$ $< p_0$ $> p_0$



c) Interpret the p-value in the context of the problem.